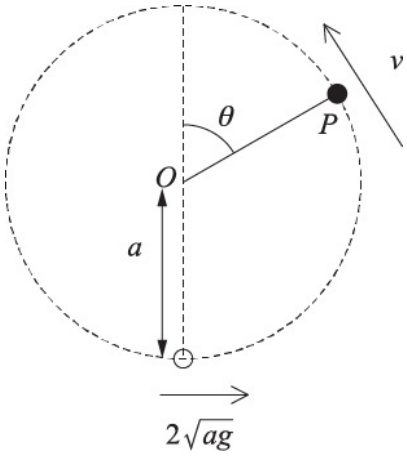
1



One end of a light inextensible string of length a is attached to a fixed point O. A particle P of mass m is attached to the other end of the string and hangs at rest. P is then projected horizontally from this position with speed $2\sqrt{ag}$. When the string makes an angle θ with the upward vertical P has speed v (see diagram). The tension in the string is T.

(i) Find an expression for T in terms of m, g and θ, and hence find the height of P above its initial level when the string becomes slack.

[6]

P is now projected horizontally from the same initial position with speed U.

(ii) Find the set of values of U for which the string does not remain taut in the subsequent motion.

[5]

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2 A simple pendulum consists of a light inextensible string of length 0.8 m and a particle P of mass m kg. The pendulum is hanging vertically at rest from a fixed point O when P is given a horizontal velocity of 0.3 m s^{-1}.

(i) Show that, in the subsequent motion, the maximum angle between the string and the downward vertical is 0.107 radians, correct to 3 significant figures.

[3]

(ii) Show that the motion may be modelled as simple harmonic motion, and find the period of this motion.

[5]

(iii) Find the time after the start of the motion when the velocity of the particle is first –0.2 m s^{-1} and find the angular displacement of OP from the downward vertical at this time.

[6]

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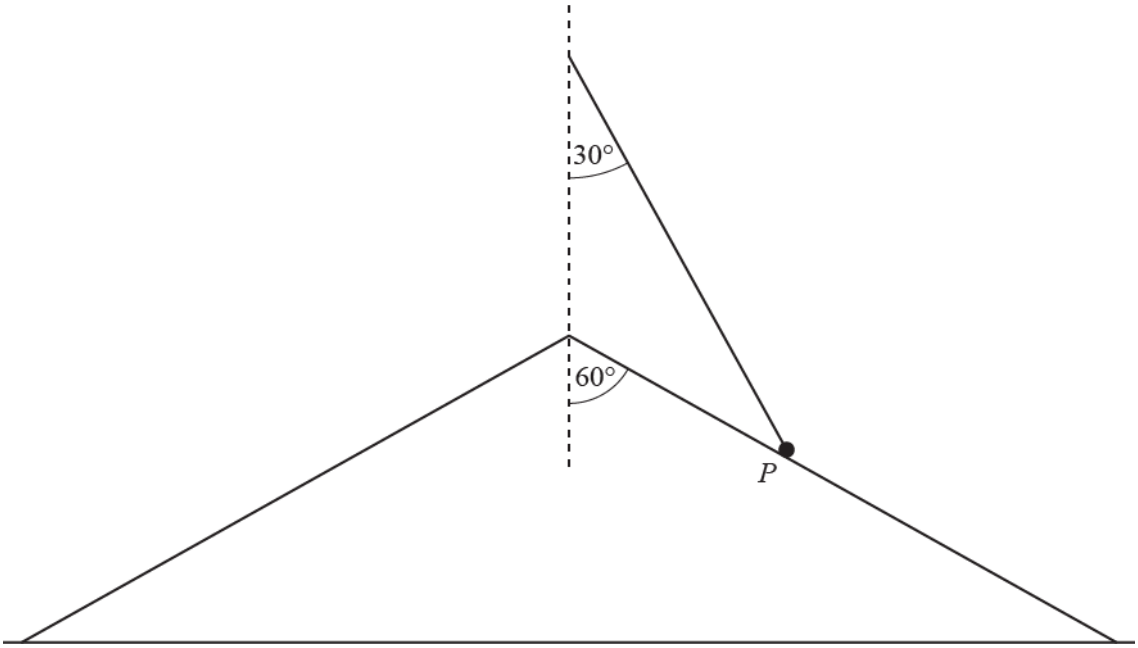

A smooth solid cone of semi-vertical angle 60° is fixed to the ground with its axis vertical. A particle P of mass m is attached to one end of a light inextensible string of length a. The other end of the string is attached to a fixed point vertically above the vertex of the cone. P rotates in a horizontal circle on the surface of the cone with constant angular velocity ω. The string is inclined to the downward vertical at an angle of 30° (see diagram).

(i) Show that the magnitude of the contact force between the cone and the particle is $\frac{1}{2}m(2\sqrt{3}g-3a\omega^2)$. **[6]**

(ii) Given that $a = 0.5$ m and $m = 3.5$ kg, find, in either order, the greatest speed for which the particle remains in contact with the cone and the corresponding tension in the string. **[3]**

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4 A particle P of mass 2.4 kg is attached to one end of a light inextensible string of length 1.4 m. The other end of the string is attached to a fixed point O on a smooth horizontal table. P moves on the table at constant speed along a circular path with O at its centre. The magnitude of the tension in the string is 21N.

(a) **(i)** Find the magnitude of the acceleration of P. **[2]**

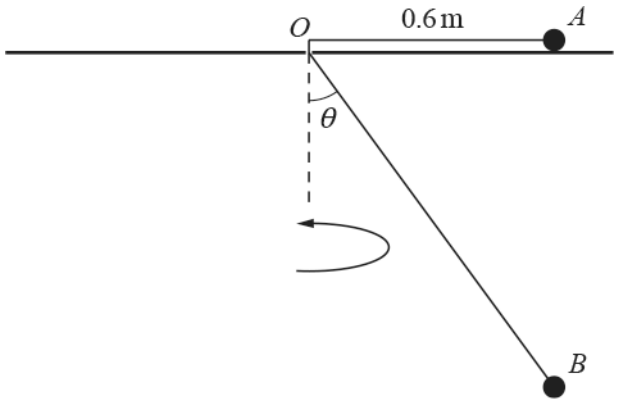
(ii) State the direction of the acceleration of P. **[1]**

(b) Find the speed of P. **[2]**

(c) Find the time taken for P to complete a single revolution. [2]

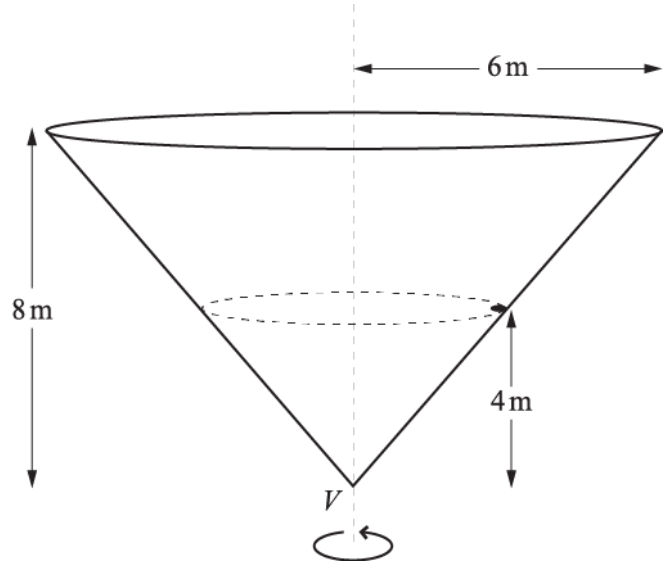
5 Two particles A and B are connected by a light inextensible string. Particle A has mass 1.2 kg and moves on a smooth horizontal table in a circular path of radius 0.6 m and centre O. The string passes through a small smooth hole at O. Particle B moves in a horizontal circle in such a way that it is always vertically below A. The angle that the portion of the string below the table makes with downwards vertical through

O is θ, where $\cos\theta = \frac{4}{5}$ (see diagram).



(a) Find the time taken for the particles to perform a complete revolution. **[7]**

(b) Find the mass of *B*. **[3]**



A conical shell has radius 6 m and height 8 m. The shell, with its vertex V downwards, is rotating about its vertical axis. A particle, of mass 0.4 kg, is in contact with the rough inner surface of the shell. The particle is 4 m above the level of V (see diagram). The particle and shell rotate with the same constant angular speed. The coefficient of friction between the particle and the shell is μ.

(i) The frictional force on the particle is FN, and the normal force of the shell on the particle is RN. It is given that the speed of the particle is 4.5 ms^{-1}, which is the smallest possible speed for the particle not to slip.

(a) By resolving vertically, show that $4F + 3R = 19.6$.

[2]

(b) By finding another equation connecting F and R, find the values of F and R and show that $\mu = 0.336$, correct to 3 significant figures.

[6]

(ii) Find the largest possible angular speed of the shell for which the particle does not slip.

[6]

END OF QUESTION PAPER

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Question			Answer/Indicative content	Marks	Part marks and guidance	
1		i	By energy	M1	Need correct 3 terms; allow	
		i	$\frac{1}{2}m(4ag) = \frac{1}{2}mv^2 + mga(1+\cos\theta)$ Use of $F = ma$	A1	wrong sign, missing/extra g, missing m / a; sin for cos correct Need 3 terms and g, allow sign	$v^2 = 2ag - 2ag\cos\theta$
		i	$T + mg\cos\theta = \frac{mv^2}{a}$	M1	slip, sin for cos	
		i	$T = 2mg - 3mg\cos\theta$	A1		
		i	Slack when $\cos\theta = \frac{2}{3}$	M1	Ft if their T has right form	
		i	Height is $\frac{5}{3}a$	A1		Examiner's Comments This part proved straightforward for most candidates, though some made sign errors and some failed to include the component of the weight term when finding an expression for T. A small number of candidates left the answer as θ, or cos θ, failing to find the height required.
		ii	If $\theta > \pi/2$, $\frac{1}{2}mU^2 > mga$	M1	Allow '=' for all M marks	
		ii	$U > \sqrt{(2ag)}$	A1	Allow ≥	
		ii	For no complete revolutions $\frac{1}{2}mU^2 < \frac{1}{2}mu^2 + 2mga$	M1	Allow wrong sign	u is vel at top
		ii	and $mg = m\frac{u^2}{a}$	M1	Allow wrong sign	
		ii	$U < \sqrt{(5ag)}$	A1	Allow ≤	$\sqrt{2ag} < U < \sqrt{5ag}$
		ii	OR Use $\frac{1}{2}mU^2 = \frac{1}{2}mv^2 + mga(1+\cos\theta)$			
		ii	and $T + mg\cos\theta = \frac{mv^2}{a}$	(M1)		
		ii	To get $T = m\frac{U^2}{a} - 2mg - 3mg\cos\theta$ oe	(A1)		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	When T = 0, $U^2 = 2ag + 3ag\cos\theta$	(M1)	Allow ≤,<	
		ii	($\theta = 0$) gives $U < \sqrt{(5ag)}$	(A1)	Allow ≤	
		ii	$(\theta = \frac{\pi}{2})$ gives $U > \sqrt{(2ag)}$	(A1)	Allow ≥	Examiner's Comments Many candidates tackled this by finding the tension in terms of *U* for a general position, and then putting it equal to zero. However, the majority using this method failed to progress any further, and those that did almost always only found the greatest possible value of *U* for the string to not remain taut in the subsequent motion. Similarly, other candidates who looked at the specific position where the string went from being taut to not being taut (or vice versa) almost only considered the topmost point on the motion and ignored the fact that the string will always be taut if the angle it makes with the upward vertical is more than 90°.
			Total	11		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
2		i	Use of energy equation	M1	Examiner's Comments This was done well by many candidates.	Allow M1 if sign error and / or 9.8 missing and / or missing *m* or *l*
		i	$0.5\ m\ (0.3)^2 = m \times 9.8 \times 0.8 \times (1 - \cos\theta)$	A1		
		i	$\theta = 0.107$	A1	No errors **AG**	0.107194171
		ii	Use $F = ma$	M1	$m \times 9.8 \sin < 9 = - m \times 0.8\ \ddot{\theta}$ Examiner's Comments Some candidates missed out the length of the pendulum, and a few others only quoted the SHM result; a few failed to work out the period of the motion.	allow M1 if sign error, or 9.8 missing
		ii	$\ddot{\theta} = -12.25\ \theta$	A1		Allow fraction
		ii	small θ	B1	Dep on having seen acc = $k\sin\theta$	Rigorous
		ii	Use of $T = \frac{2\pi}{\omega}$	M1	or sight of $\omega = 3.5$	
		ii	$T = 1.80$	A1		accept $\frac{4\pi}{7}$ (1.795195)

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	identifying amplitude as 0.107	B1	Examiner's Comments This part proved more challenging, though most candidates did tackle it by means of the equations for velocity and displacement in terms of cos ωt and sin ωt. Those who chose to use $\theta^2 = \omega^2(\alpha^2 - \theta^2)$ were able to find the angular displacement but were not usually able to correctly find the time, since this required some manipulation involving the period of motion. For both methods, some candidates used a linear speed of –0.2 m s^{-1} when they should have used an angular speed of –0.25 rad s^{-1}.	ft from (i)
		iii	Use of $(\dot{\theta}) = 0.107 \times 3.5 \times \cos(3.5t)$	M1	or sin (3.5t + ε), ε not 0	ft for a and ω; allow sign error
		iii	Use of $\dot{\theta} = -0.25$	A1	Consistent angle or length	
		iii	$t = 0.658$	A1		(0.6576339)
		iii	Use of $\theta = 0.107 \sin(3.5t)$	M1	ft from velocity equation (matches, ignore sign)	
		iii	(θ =) 0.0797rads	A1	accept 5.20°	(0.0796678 or 0.079576)
			Total	14		

Question			Answer/Indicative content	Marks	Part marks and guidance	
3		i		M1*	Resolving vertically (3 terms)	
		i	$T\cos 30 + R\sin 60 = mg$	A1		
		i		M1*	Resolving horizontally (3 terms); an r used where r is not just a	
		i	$T\sin 30 - R\cos 60 = m(a\sin 30)\omega^2$	A1		
		i		M1 dep*	Eliminating T and solve for R in terms of m, g, a and ω	
		i	$R = \frac{1}{6}m\left(2\sqrt{3}g - 3a\omega^2\right)$	A1	**AG** Correctly shown **Examiner's Comments** The usual method of resolving parallel to and perpendicular to the direction of the acceleration was successfully employed by the majority of candidates. However some used the wrong radius in their acceleration and others found the algebraic manipulation to achieve the given answer a challenge. A minority of candidates attempted to resolve in other directions with not much success, failing to realise that the acceleration has a component in the direction they were resolving.	
		ii	For using $R = 0$ to attempt to find either v or T	M1	Or attempt to find ω	
		ii	$T = \frac{mg}{\cos 30} = 39.6$	A1	39.606228…	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\omega^2 = \frac{T}{ma}$, $v = 1.19\ \text{m s}^{-1}$	A1	1.1893309… Examiner's Comments Most candidates were aware that using $R = 0$ was the way to proceed in this question. However two requests were made with a significant number not attempting to find the tension in the string. In other cases, candidates only attempted the angular speed despite the request being for the speed of the particle.	
			Total	9		

Question			Answer/Indicative content	Marks	Part marks and guidance	
4		a	(i) 21 = 2.4*a* 8.75 m s^{-2} (ii) Towards *O*	M1(AO 1.1a) A1(AO 1.1) [2] B1(AO 1.1) [1]	Use of Newton II	
		b	$8.75 = \frac{v^2}{1.4}$ 3.5 m s^{-1}	M1(AO 1.1a) A1(AO 1.1) [2]	Equating $\frac{v^2}{r}$ to their value from (a)	
		c	$\frac{2\pi \times 1.4}{3.5}$ 2.51 s	M1(AO 1.1a) A1(AO 1.1) [2]	$2\pi r$ divided by their value from (b)	
			Total	7		

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Question			Answer/Indicative content	Marks	Part marks and guidance		
5		a	$T\cos\theta = m_B g$ $T\sin\theta = m_B \times 0.6 \times \omega^2$ $\tan\theta = (0.6\omega^2) / g$ $\tan\theta = ¾$ oe $\omega = 3.5$ $t = \frac{2\pi}{3.5}$ Time for one revolution is 1.8 seconds	M1* (AO3.1b) M1* (AO3.3) M1dep (AO3.1b) B1 (AO1.1) A1 (AO1.1) M1 (AO1.1) A1 (AO3.2a) [7]	Balancing vertical forces on *B* NII for *B* with $r = 0.6$ (could use $v^2 / 0.6$) Combining equations and eliminating *T* May be implied. Accept θ = 36.9 Their 3.5	Or $t = 2\pi r/v$ ($v = 2.1\,m/s$)	

<u>Examiner's Comments</u>

This question was about two connected particles moving in horizontal circles with one forming a conical pendulum. Candidates needed to focus on particle B, resolving forces vertically and horizontally using Newton's Law's and the formula for radial acceleration.

This item was challenging for most candidates, who often just wrote any formulas that they thought might be relevant. More successful candidates were able to resolve the tension in the string at B correctly to form suitable equations that could then be solved to find ω and t.

Less successful candidates were often distracted by

Question			Answer/Indicative content	Marks	Part marks and guidance	
					particle A and tried for example to relate the tension at A to the tension at B in some way(which was the intention of part (b)). Other common problems included resolving in the wrong direction with candidates frequently stating $T = mg\cos\theta$ rather than $T\cos\theta = mg$, and assuming that they needed to find T in order to find ω. Many candidates did not recognise the use of a 3, 4, 5 triangle, which gives simple exact values of $\sin\theta$ and $\tan\theta$, opting instead to use $\theta = 36.9°$. Practice of finding the corresponding $\sin\theta$ and $\tan\theta$ expressions from a given $\cos\theta = a/b$ using a sketch and Pythagoras' Theorem (and the equivalents from a given $\sin\theta$ or $\tan\theta$ value), will help avoid complex decimal coefficients in algebraic expressions and ensure access to exact answers when required. Exemplar 4	

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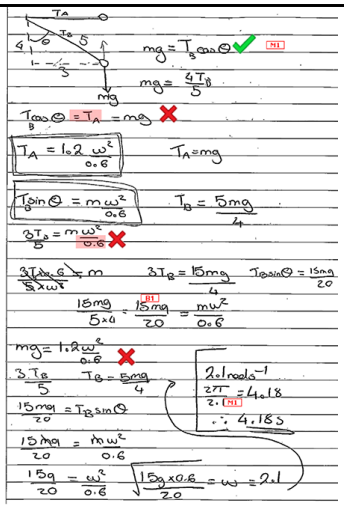
Question			Answer/Indicative content	Marks	Part marks and guidance	
					 In this exemplar, the candidate has attempted a variety of approaches, some correct and some incorrect. For example, at one point they have equated the weight of particle B with the tension at A. We have given them credit wherever possible. For resolving *T* vertically, i.e. $T\cos\theta = mg$, they have gained M1. Unfortunately, they have used $a = \omega^2/r$ instead of $r\omega^2$ or v^2/r when resolving horizontally, and therefore do not have two suitable equations to solve. Notice however, that they have also written *T* as $mg/\cos\theta$ and multiplied by $\sin\theta$ to get ¾*mg* (= $mg\tan\theta$), seen as 15/20. We have taken this to imply that $\tan\theta$ = ¾ and therefore given B1 here. Later on, they have produced a value for ω, albeit from wrong equations, but gain M1 for correctly using it to get a value of *t*.	

Question			Answer/Indicative content	Marks	Part marks and guidance		
		b	$T = 1.2\times0.6\omega^2$ (= 8.82) $8.82\cos\theta = m_B g$ or $8.82\sin\theta = m_B\times0.6\omega^2$ $m_B = 0.72$	M1 (AO2.2a) M1 (AO3.1b) A1 (AO1.1) [3]	NII for A and for realising that ω is the same for A and B. Substituting their T into either of their equations of motion for B. <u>Examiner's Comments</u> This question required candidates to equate the tension at B with the tension at A in order to determine the mass of particle B. It also required rearranging one of their equations from the start of part (a) and using their value of ω or v. Most candidates who had successfully completed part (a) went on to complete part (b) although some made errors, such as assuming that the centripetal force at A was the same as at B.	Could be seen in (a)	
			Total	10			

Question			Answer/Indicative content	Marks	Part marks and guidance	
6		i	(a) 0.8F + 0.6R = 0.4g	M1	Attempt to resolve vertically	
		i	4F + 3R = 19.6 **[AG]**	A1	www Examiner's Comments Some good solutions were seen. Some candidates either got a wrong angle or got a correct angle and resolved incorrectly, but amazingly still managed to get the correct result.	
		i		M1	Attempt with three terms.	
		i	(b) 0.8R – 0.6F = 0.4 x $4.5^2/3$	A1	aef including cos, sin correct angle	
		i	Solve for R or F	M1	Use 2 relevant resolutions.	
		i	F = 1.516	A1		
		i	R = 4.512	A1		
		i	Use μ= F/R to get μ = 0.336 **[AG]**	B1	Examiner's Comments Finding a second appropriate equation proved difficult for quite a few. A remarkable number of combinations of values of *F* and *R* appear to produce μ = 0.336! Some candidates lost marks here by eliminating both *R* and *F* from their equations to find just μ and so failing to find the values of *R* and *F* as required.	
		ii		M1	Resolve vertically, three terms	
		ii	0.6R – 0.8F = 0.4g	A1		
		ii	R = 11.8 or F = 3.98	A1		
		ii		M1	N2L, resolve horizontally, three	

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$0.8R + 0.6F = 0.4 \times 3\ \omega^2$	A1	terms	
		ii	$\omega = 3.14$ rad s^{-1}	A1	Examiner's Comments This proved to be the most difficult question on the paper with very few correct solutions seen by examiners. Only a minority realised the direction of *F* had to change in this situation and most simply used the same equations as before, or used one equation with the values of *R* and *F* found in (ii), or solved equations in which *F* (or *R*) was zero.	
			Total	14		

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